INCH-POUND

MIL-G-0043914E(GL)
26 November 1990
USED IN LIEU OF
MIL-G-43914D
9 January 1985

MILITARY SPECIFICATION

GOGGLES, SUN, WIND AND DUST: 1974

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-G-43914D. It is subject to modification. However, pending it's promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers two types of goggles for wear with and without glasses.

1.2 Classification. The goggles shall be of the following types and classes as specified (see 6.2).

Type I - For wear without face mask.
Class 1 - Deleted (see 6.8)
Class 2 - Deleted (see 6.8)
Class 3 - Clear (ballistic)
Class 4 - Neutral gray (ballistic)
Class 5 - Light laser protective (ballistic)
Class 6 - Dark laser protective (ballistic)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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Type II - For wear with face mask.
Class 3 - Clear (ballistic)
Class 4 - Neutral gray (ballistic)
Class 5 - Light laser protective (ballistic)
Class 6 - Dark laser protective (ballistic)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
L-P-393	-	Plastic Molding Material, Polycarbonate, Injection and Extrusion
JJ-W-155	-	Webbing, Textile, (Cotton, Elastic)
QQ-B-613	-	Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet and Strip)
CCC-D-950	-	Dyeing and Aftertreating Processes for Cotton Cloths
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-665	-	Boxes: Paperboard, Metal Edged and Components

MILITARY

MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base
MIL-F-21840	-	Fastener Tapes, Hook and Loop Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-V-43511	-	Visors, Flyer's Helmet, Polycarbonate
MIL-H-46170	-	Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hyprocarbon Base

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-810	-	Solar Radiation, Methods of Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-2-635	-	Goggles, Sun, Wind and Dust: 1974 (Type I)
8-2-636	-	Head Band Assy. (Type I and II Goggles)
8-2-637	-	Frame (Type I Goggles)
8-2-639	-	Foam Pad (Type I and II Goggles)
8-2-640	-	Flannelette Cloth, Top (Type I and II Goggles)
8-2-641	-	Flannelette Cloth, Bottom (Type I and II Goggles)
8-2-664	-	Lens (ballistic), Class 3 and Class 4
8-2-827	-	Goggles, Sun, Wind and Dust: 1974 (Type II)
8-2-828	-	Frame (Type II Goggles)
8-2-829	-	Fastener, (Hook), Upper and Lower

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2000 - Rubber Products in Automobile Applications
D 3951 - Standard Practice for Commercial Packaging

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-z80.1-1979 - Lenses, Recommendations For Prescription Opthalmic

(Application for copies of ANSI publications should be addressed to ANSI, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Rubber. The frame for the goggles shall be molded of a rubber compound having polychloroprene as a base material and shall meet the requirements of type, class, grade and suffix letters, M3BC 710 A14 G21 of ASTM D 2000.

3.2.2 Lenses. The material for the ballistic lenses shall be fabricated from clear, neutral gray, light laser, or dark laser bisphenol type polycarbonate plastic molding material conforming to L-P-393.

3.2.2.1 Class 3 and 4 lenses. The class 3 and 4 ballistic lenses shall have a thickness of 0.080 $\pm$ 0.001 (see Drawing 8-2-664), shall have an abrasion resistant coating, and shall conform to class 1 and 2, respectively, of MIL-V-43511 with the following exceptions:

a. Design and dimensions. The design and dimensions of the lenses shall be in accordance with Drawing 8-2-664.

b. Areas of vision. The lens has three areas of vision which are identified as: outside the area of vision (no optical requirement), non-critical area of vision, and critical area of vision. The center point for the left and right optics shall be identified as point "C" and the location and size of the three areas of the lens shall be as specified in Drawing 8-2-664. The numbered circles within the critical areas are designated as points of choice for

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prismatic and distortion tests. Points bearing the same number, for example 5R and 5L, shall be compared with each other, except points 2R and 2L (outermost point on left and right sides of lens) shall be compared with point 3 (center point of lens) when measuring refractive power of prismatic deviations.

3.2.2.2 Class 5 and 6 lenses. The class 5 and 6 lenses shall have a thickness of 0.080 ± 0.001 (see Drawing 8-2-664), shall have an abrasion resistant coating, and shall conform to class 1 of MIL-V-43511 with the following additions and exceptions:

a. Design and dimensions. The design and dimensions of the lenses shall be in accordance with Drawing 8-2-664.

b. Areas of vision. The lens has three areas of vision which are identified as: outside the area of vision (no optical requirement), non-critical area of vision, and critical area of vision. The center point for the left and right optics shall be identified as point "C" and the location and size of the three areas of the lens shall be as specified in Drawing 8-2-664. The numbered circles within the critical areas are designated as points of choice for prismatic and distortion tests. Points bearing the same number, for example 5R and 5L, shall be compared with each other, except points 2R and 2L (outermost point on left and right sides of lens) shall be compared with point 3 (center point of lens) when measuring refractive power or prismatic deviations.

c. Optical density (class 5). The lens shall incorporate absorptive dyes in a concentration to yield an optical density (OD) of 4.0 or greater at 694.3 nm (ruby laser wavelength) and 1064 nm (neodymium laser wavelength) for laser radiation incident normal to the substrate with any polarization state when tested in accordance with 4.5.4.1.

d. Optical density (class 6). The lens shall incorporate absorptive dyes (see 6.6) in a concentration to yield an optical density (OD) of 4.0 or greater at 532 nm (double neodymium laser wavelength), 694.3 nm (ruby laser wavelength), and 1064 nm (neodymium laser wavelength) for laser radiation incident normal to the substrate with any polarization state when tested in accordance with 4.5.4.1.

e. High energy laser. The lens shall meet the optical density requirements when tested against laser radiation at the energy or power densities at 532 (class 6), 694.3 and 1064 nm (class 5 and 6) when tested in accordance with 4.5.4.2.

f. Luminous transmittance. The scotopic luminous transmittance (for dark adapted eye), assuming CIE standard Illuminant C, shall not be less than 0.45 (class 5), 0.09 (class 6), when tested in accordance with 4.5.4.3.

g. P43 transmittance. The photopic luminous transmittance (for the light adapted eye) shall not be less than 40 percent (class 5) and 13 percent (class 6) when the source spectral emission is that of the P43 phosphor when tested in accordance with 4.5.4.4.

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h. Neutrality. The neutrality of the class 5 and 6 lenses shall not be measured.

i. Chromaticity. The chromaticity coordinates of the class 5 and 6 lenses shall not be measured.

j. Chemical resistance. The lens shall be resistant to attack from chemicals including but not limited to the following list when tested as specified in 4.5.5.1. The optical density requirements and transmittance requirements of the lens shall be met following the chemical resistance test.

DEET insect repellent
(Diethyl meta toluamides in ethanol)
Decontamination fluid
Fire resistant hydraulic fluid (FRH-MIL-H-46170)
Hydraulic Fluid, Petroleum Base (MIL-H-6083)
Gasoline
Motor Oil
JP4 Jet Fuel
JP8 Multi-Purpose Fuel

k. Temperature. The lens shall meet the requirements for optical density and luminous transmittance following exposure for 72 hours at 160°F and 72 hours at -60°F +3°F when tested in accordance with 4.5.5.2.

l. Solar radiation. The lens shall meet the requirements for optical density and luminous transmittance after being tested in accordance with 4.5.5.3.

m. Humidity. The lens shall meet the requirements for optical density and luminous transmittance after exposure to a combined total of 240 hours of temperature and humidity when tested in accordance with 4.5.5.4.

3.2.3 Webbing. The elastic webbing for the headband shall conform to type I, class 3 of JJ-W-155, except that the webbing shall be mildew resistant treated and the fungicide content shall be 0.5 percent to 1 percent based on the cotton component of the webbing only. The color of the webbing shall be Olive Drab 7.

3.2.3.1 Dipping compound. The ends of elastic webbing for the headband shall be dipped in a cellulose acetate or cellulose acetate butyrate in suitable organic solvent having a concentration of 25 to 28 percent solids to a depth of 5/8 ± 1/8 inch and allowed to dry (see 6.6).

3.2.4 Cloth. The cloth for the filtering screen shall be a commercial cotton flannelette dyed black with direct colors and napped on one side only. The cloth shall have a minimum weight of 4.0 ounces per square yard and a minimum texture of 42 ends per inch and 40 picks per inch when tested as specified in 4.4.1.2.

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3.2.4.1 Mildew resistance. The mildew resistant treatment of the cotton flannelette shall conform to CCC-D-950, except that the inhibitor "a" fungicide content shall be 0.5 percent to 1.0 percent and shall meet the requirements applied to inhibitor "a", and that the soil burial test shall not apply.

3.2.5 Adhesive for attaching flannelette. The adhesive used in attaching the flannelette to the rubber frame of the goggles shall be 3M's Adhesive EC 1300L or equal. The adhesive shall be applied in such a manner that the flannelette will not peel when tested for peel strength as specified in 4.4.7.

3.2.6 Foam pad. The foam padding on the goggles shall be a 3/8 inch thick polyether polyurethane foam, open cell, having a density of 1.40 $\pm$ 0.20 pounds per cubic foot. The foam shall be gray in color, and possess nominal 55 pores per inch with a minimum of 35 pores per inch and a maximum of 75 pores per inch. A black polyurethane skin, 2 mils thick, shall be adhered to one side of the foam only. The skin shall not be capable of being peeled without tearing the foam when tested as specified in 4.4.1.2 (see 6.5).

3.2.6.1 Adhesive for attaching foam. The adhesive used in attaching the foam to the goggles shall be 3M's Adhesive EC 1300L or equal. The adhesive shall be applied in such a manner that the foam cannot be separated from the goggles without tearing the foam when tested as specified in 4.4.7.

3.2.7 Metal components.

3.2.7.1 Snap fasteners. The snap fasteners shall be brass, any composition conforming to QQ-B-613. The following components of the snap fastener shall be United-Carr, Inc., number or equal: Button No. 21154, 16 line size; Socket No. 22201; Stud No. 22303; Eyelet No. 22402.

3.2.7.2 Slide, 3/4 inch. The slides shall be hard brass conforming to QQ-B-613, any composition and shall conform to Waterbury Buckle Company No. 164, or equal.

3.2.7.3 Finish. All metal parts shall be given a black chemical finish conforming to MIL-F-495.

3.2.8 Tape, hook fastener (for type II goggles). The hook fastener tape for the type II goggles shall conform to type I, class 2, color green 3421 of MIL-F-21840 and shall conform to Drawing 8-2-829. Two strips of the hook tape shall be stitched together lengthwise to form a sufficiently wide hook strip from which to cut the piece to be adhered to the lower part of the goggles.

3.2.8.1 Adhesive for attaching the hook fastener tape. The adhesive to be used in attaching the hook fastener tape pieces to the goggle frame shall be 3M's adhesive EC 1300 L or equal. The adhesive shall be applied in such a manner so that there is no delamination of more than 1/8 inch when tested as specified in 4.4.7 (see 6.4).

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3.3 Design and construction.

3.3.1 Design. The goggles shall be designed to provide binocular vision, to hold the interior of the lens a sufficient distance from the eyes to allow adequate clearance for eyeglasses, and to provide circulation of air to the interior of the goggles. The goggles shall be fitted with a urethane foam pad in the area of contact with the face and cloth filtering screens over ventilating holes. The type II goggle frame shall have extensions in the nose, cheek and forehead areas to which fastener tape pieces have been adhered as shown on Drawing 8-2-827.

3.3.2 Construction. The goggles shall be constructed as shown on drawings listed in 2.1.2. The napped side of the filtering screen cloth conforming to 3.2.4 shall be cemented to the frame with the adhesive specified in 3.2.5. The foam pad conforming to 3.2.6 shall be cemented to the frame using the adhesive specified in 3.2.6.1. The foam pad shall be cemented on the non-skin inside (see section A-A in Drawing 8-2-635 and Drawing 8-2-827) of the rubber frame. The width of the urethane foam shall correspond within $\pm$ 1/8 inch to the width of the rubber facing. The open area required for the temple when spectacles are worn with the goggles shall be fully covered with the urethane foam as shown on Drawings 8-2-635 and 8-2-827. The urethane foam shall not extend into the area of vision. The ends of the webbing for the headband (see 3.2.3), shall be dipped in resin conforming to 3.2.3.1 to prevent unraveling. The hook fastener tape pieces for the type II goggles shall be adhered as shown on Drawing 8-2-827 with the adhesive specified in 3.2.8.1.

3.3.3 Goggles complete.

3.3.3.1 Type I. The type I goggles shall be furnished with class 3 and 4 lenses or class 5 and 6 lenses as specified (see 6.2). A clear lens shall be inserted in the frame.

3.3.3.2 Type II. The type II goggles shall be furnished with class 3 and 4 lenses or class 5 and 6 lenses as specified (see 6.2). The goggles shall be furnished without a lens inserted in the frame.

3.4 Marking. Each frame shall be permanently and distinctly marked with the information shown in Drawing 8-2-637, in the location shown, and in accordance with MIL-STD-130.

3.5 Workmanship. The frame, lenses and the respective components shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor

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e examined " 5.1". Shipping container examination. Shipping containers shall be

may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall consist of a lot of 1000 complete systems. Of these, 75 systems shall be submitted to the indicated locations for the tests in table I. In addition, the first article shall be examined for the defects specified in 4.4.4, 4.4.5 and 4.4.6. A sample size of 75 systems shall be used when testing in 4.5.4, and a sample size of 32 systems shall be used when testing in 4.5.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

TABLE I. First article tests

Requirement	Requirement paragraph	Test paragraph	Test locations	Quantity
High energy laser and transmittance	3.2.2.2	4.5.4.1 4.5.4.2	USA Environmental Health Agency (AEHA)	25

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be hermetically sealed, by evacuating the air within a sealing chamber under a

TABLE I. First article tests (cont'd)

Requirement	Requirement paragraph	Test paragraph	Test locations	Quantity
High energy laser and transmittance (cont'd)		4.5.4.3 4.5.4.4	Letterman Army Institute for Research (LAIR)	25
Environmental	3.2.2.2	4.5.5.1 4.5.5.2 4.5.5.3 4.5.5.4	AEHA	25

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be accepted as evidence that components and materials listed below conform to the requirements in listed paragraphs.

Component	Characteristic	Requirement
Rubber	Composition	3.2.1
Plastic	Composition	3.2.2
Dipping compound	Composition	3.2.3.1
Cloth, cotton flannelette	Dyed with direct colors	3.2.4
	Napped one side only	3.2.4
Adhesive for flannelette	Composition	3.2.5
Foam pad	Composition, density	3.2.6
Adhesive for foam pad	Composition	3.2.6.1
Snap fastener	Material and finish	3.2.7.1 and 3.2.7.3

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Slide	Material and finish	3.2.7.2 and 3.2.7.3
Adhesive for fastener hook tape	Composition	3.2.8.1

4.4.1.2 Component testing. In addition to any testing required by 4.4.1, the components shall be tested for the applicable characteristics in table II. For the flannelette, the lot size shall be expressed in yards, the sample unit shall be 1 yard, and the inspection level shall be S-1. For the foam skin, the sample shall consist of one specimen selected at random from the lot. Any test failure shall be cause for rejection of the lot. All test reports shall contain the individual values utilized in expressing the final results.

TABLE II. Component tests

Characteristic	Requirement paragraph	Test method
Flannelette weight	3.2.4	5041 of FED-STD-191
Flannelette texture	3.2.4	5050 of FED-STD-191
Foam skin peel strength	3.2.6	1/

1/ Peel the skin from the foam by hand. If the skin separates without tearing the foam, it shall be considered a test failure.

4.4.2 Lens inspection (classes 3 and 4). The lenses shall be inspected for conformance to the requirements in 3.2.2.1 in accordance with the end item examination and testing provisions of MIL-V-43511 except that the unit inspected shall be a lens.

4.4.3 Lens inspection (classes 5 and 6). The lenses shall be inspected for conformance to the requirements in 3.2.2.2 in accordance with 4.4.4, 4.4.5 and 4.4.6.

4.4.4 End item visual examination. The end items shall be examined for the applicable defects in table III. The lot size shall be expressed in units of goggles of one type only. The sample unit shall be one pair of goggles with applicable lenses. The inspection level shall be II and the acceptable quality levels (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Rubber face mask	Crack extending through body	101	
	Surface crack not extending through body		201

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Rubber face mase (cont'd)	Blister or contact scar 1/4 inch or over in largest dimension	102	
	Blister or contact scar over 1/64 inch but less than 1/4 inch in largest dimension		202
	Rough surface, e.g., multiple small lumps or blisters 1/	103	203
	Rubber not solid, i.e., any undesirable sponginess or porosity 1/	104	204
	Excessive flash or poor trim		205
	Not molded in one piece	105	
	Lump, void, or abrasion 1/	106	206
	Any cut, puncture, sharp crease, wrinkle, or tear	107	
	Any hole (except vents) more than 1/16 inch in largest dimension	108	
	More than two pinholes, 1/16 inch or less in largest dimension, in any one square inch of surface area		207
	Objectionable odor (other than typical rubber odor)		208
Fastener hook tape (type II only)	Any delaminations	109	
	Crushed hooks, area 1/4 inch or less in diameter:		
	One area		209
	More than one area	110	
	Crushed hooks, area more than 1/4 inch in diameter, one or more areas	111	
	Improperly located on frame:		
	1/16 to 1/8 inch		210
	More than 1/8 inch	112	
	Excessive adhesive affecting appearance		211
	Excessive adhesive on hooks	113	
Webbing	Color not as specified		212
	Color offshade, mottled, or streaky		213
	Clearly noticeable crystallization of mildew inhibitor 1/	114	
	Cut, hole, or tear	115	
	Frayed or scalloped edge; not tightly or firmly woven 1/	116	
	Cut ends of webbing not dipped in dipping compound	117	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Metal components (snap fasteners and slides)	Not specified type, size, or design	118	
	Broken, cracked, fractured, damaged, malformed, bent out of shape, or otherwise impaired 1/	119	
	Evidence of corrosion	120	
	Finish flakes or rubs off 1/	121	
Design and construction	Varies from design or construction shown on specified drawing	122	
	Any characteristic not in accordance with specified design, e.g. not designed to accommodate plastic lens or provide binocular vision, adequate eyeglass clearance or circulation of air to interior of goggles	123	
	Any component missing	124	
Workmanship and assembly	Frame malformed, bent out of shape or otherwise impaired 1/	125	214
	Size or shape of ventilation holes not as specified		215
	Ventilator hole missing		216
	Row of ventilator holes missing	126	
	Ventilator holes not spaced as shown on drawing 1/	127	217
	One to three ventilator holes not covered (nor adhered by flannelette)		218
	Four or more ventilator holes not covered (nor adhered by flannelette)	128	
	One to three ventilator holes filled with adhesive		219
	Four or more ventilator holes filled with adhesive	129	
	Snap fastener not properly located on tab as shown on drawing	130	
	Snap fastener loose or not properly clinched to tab	131	
	Urethane foam cemented on skin side	132	
	Urethane foam extending beyond outer periphery of rubber facing more than 1/8 up to 1/4 inch		220
	Urethane foam extending beyond outer periphery of rubber facing by more than 1/4 inch	133	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship and assembly (cont'd)	Urethane foam extending beyond inner periphery of rubber facing by more than 1/8 inch	134	
	Urethane foam recessed from inner or outer periphery of rubber facing by more than 1/8 inch		221
	Any one area of non-adherence of foam pad 1/4 square inch up to but not including 1/2 square inch		222
	Any one area of non-adherence of foam pad 1/2 square inch or greater	135	
Marking	Missing, incomplete, illegible, misspelled, not specified type, not in specified location, or not applied in the specified manner		223

1/ Defects designated by a 1/ in the defect column shall be classified as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

4.4.5 Goggles dimensional examination. The goggles shall be examined for conformance to all specified dimensions. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of goggles of one type only. The sample unit shall be one pair of goggles. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.6 Goggles lens fit and assembly examination. The goggles and lenses shall be examined for the defects listed below. The lot size shall be expressed in units of goggles. The sample unit shall be one pair of goggles with applicable lenses. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Lens fit	Rough edge on lens from a worn die or from excess flash
	Shape of lens when inserted in frame does not conform to curvature of frame
	Lens does not fit snugly in frame
Lens assembly	Snap fastener does not function properly, i.e., fails to effect a secure closure or exceptionally difficult to open

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Lens assembly (cont'd)	Snap fastener stud or socket too tightly affixed resulting in damage to rubber Snap fastener tears plastic lens upon insertion of lens into frame Stud not long enough to hold thickness of one lens and frame

4.4.7 End item testing. The end items shall be tested for the characteristics indicated in table IV. The sample unit shall be one pair of goggles. The sample size shall be five goggles selected at random from the lot. Any test failure shall be cause for rejection of the lot.

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method	No. of determ. per sample unit	Results reported as
Peel strength of flannelette to goggles	3.2.5	4.5.1	4	Pass or fail
Peel strength of foam to goggles	3.2.6.1	4.5.2	4	Pass or fail
Peel strength of fastener hook tape to goggles (type II only)	3.2.8.1	4.5.3	2	Pass or fail

4.4.8 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling Incorrectly fabricated plastic or cotton flannel bag Bulged or distorted container

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Contents	Number of goggles per shipping container is more or less than required

4.4.8.1 Intermediate pack of separately procured lens examination. The lenses packaged for shipment shall be examined to determine conformance with unit and intermediate pack markings and specified quantity. The sample unit shall be one intermediate pack. The lot size shall be the number of intermediate packs in the inspection lot. Any intermediate pack containing less than the specified or marked quantity of lenses shall be classified as a defect. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.9 Palletization inspection. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Peel strength of flannelette to goggles. None of the four strips of flannelette shall be capable of being peeled in one piece (from the rubber) when an edge of the flannelette is pulled by hand. The attempt to peel the flannelette should result in the flannelette shredding and leaving some threads still bonded.

4.5.2 Peel strength of the foam to goggles. The foam shall not be capable of being pulled in four separate areas (away from the rubber) by hand without tearing the foam.

4.5.3 Peel strength of fastener hook tape to type II goggles. Two 1/2-inch wide by 2-inches long test specimens shall be cut from a goggle. Peel a 1-inch section of each test specimen, attach the fastener hook tape material to a suitable fixed holder and the rubber strip to a 1-pound weight, distributed over the full width of the specimen. Allow the weight to hang free for 1 hour. At

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the end of the test period, measure the distance of peel for each specimen and calculate the average to the nearest 1/16 inch. The test shall be performed at a temperature of $70^{o} \pm 2^{o}F$.

4.5.4 Optical tests.

4.5.4.1 Optical density. The optical density of the lens shall meet the specified value for a radiant exposure of 20 mJ/cm for Q-switched emissions less than 40 ns and greater than 1 ns. Laser measurement of the optical density shall be performed at the following location on the lens; 47 nm from the vertical line bisecting the lens and on the horizontal centerline (see figure 1). The 47 nm is to be measured on a line tangent to the vertices of the left and right lenses. The exposure shall be made normal to surface at the position and the beam shall be incident form the convex side of the lens. The incident irradiance diameter at the lens must be 4 nm or greater. The spatial distribution of the beam should be as near uniform as possible. The optical density of the lens shall be measured by one of the following methods:

4.5.4.1.1 Spectrophotometer. The optical density (absorbencies) of the sample may be determined with commercially available dual beam spectrophotometers capable of measuring to the level required in 3.2.2.2. The effective sensitivity of the spectrophotometer may be increased by placing a calibrated absorption filter in the reference beam. The spectral bandwidth of the spectrophotometer shall be not greater than 2 nm for visible wavelengths and no more than 25 nm for wavelengths in the near infrared.

4.5.4.1.2 Densitometer. The optical density may be measured using a suitably constructed, fixed wavelength densitometer which incorporates narrow band filters, a light source, a detector, and a readout device capable of measuring the required optical density. The bandwidth of the filters shall be no greater than 2 nm in the near infrared.

4.5.4.1.3 Laser. The optical density may be measured directly with lasers at the specified wavelengths as the light source, using procedures described in 4.5.4.3.

4.5.4.2 High energy laser. Some absorptive materials can bleach or saturate when exposed to high power densities, thus the effective optical density at a high power may be lower than the optical density measured at low power with a spectrophotometer or photometer.

4.5.4.2.1 Test facility. The test set up is shown schematically in figure 1. It includes a laser and a calibrated radiometer. The radiometer generally reads a charge in coulombs which is converted to energy with a calibration factor. Several aperatures and baffles are used to prevent scattered light from reaching the detector. A narrow band transmission filter is placed in front of the detector to block out band radiation. The sample is placed as shown and neutral density filters may be placed in the beam either in front of or behind the sample as required. A beam splitter and second radiometer are used to sample the beam and monitor the energy of each lot.

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4.5.4.2.2 Laser calibration. The total output energy of the laser is measured with a calibrated radiometer. The narrow band transmittance filter should be removed since it is often difficult to know its transmittance accurately. Calibrated neutral density filters may be used to reduce the energy to a level which will not damage the detector. The pulse width should be measured periodically with a fast photodiode and oscilloscope. The power is found by dividing the energy per pulse by the pulse width. The beam diameter D, defined as the $1/e^2$ points, assuming a gaussian profile, is found from the formula:

$$D = \sqrt{\frac{-d^2}{\ln(1 - Pd/Po)}} \quad \text{OR} \quad \left\{\frac{-d^2}{\ln(1 - Pd/Po)}\right\}^{1/2}$$

where d is the diameter of a reduced aperture, Pd is the measured power with the reduced aperture, and Po is the full power measured with an unrestricted aperture. Note, in all of the optical density measurements the full aperture is used. The energy (or power) density is found by dividing the energy (or power) by the area of the beam.

4.5.4.2.3 Data collection. With suitable neutral density (ND) filters but no sample in the beam, five shots should be fired and the charge reading in coulombs of both the direct and reference radiometers recorded to establish a base line. With the sample in the beam generally replacing some or all of the ND filters, the charge readings for three shots are recorded for the direct and referenced radiometers. An appropriate ND filter will generally be required in front of the sample to the desired level.

4.5.4.2.4 Data reduction. Let D be the optical density of the filters in the beam when the base line (no sample present) is established. Let Co be the direct radiometer reading and Cr the reference radiometer reading. Calculate the average of each, designated by Co and Cr, respectively. For the three shots with the sample in place let Cm' be the average of the measured readout of the reference radiometer. The corrected (normalized) measured charge, Cm, is then

$$Cm = \frac{\overline{Cr}}{Cr} \quad Cm' \qquad (1)$$

Let F be the optical density of any ND filters in the beam whether they are in front of or behind the sample and let S be the optical density of the sample. Then

$$10^{-D} = k\overline{Co} \qquad (2)$$

and

$$10^{-(F + S)} + kCm \qquad (3)$$

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where K is a constant. Dividing the first of these equations by the second and taking the lot of both sides results in

$$S = D - F + \text{Log}(\bar{C}o/Cm) \qquad (4)$$

Equation 4 is used to calculate the effective optical density of the sample. If B is the optical density of the filter in front of the sample, the energy density incident on the sample is given by

$$E = Eo \times 10^{-B}$$

where Eo is the direct output energy density of the laser.

4.5.4.3 Luminous transmittance. The scotopic luminous transmittance of the LPF shall be measured with a spectrophotometer at 10 nm intervals in the wavelength range of 380 nm to 780 nm. The scoptic luminous transmittance is calculated according to the methods described in 4.5.4.3.1 and 4.5.4.3.2.

4.5.4.3.1 Definitions. Photopic and scotopic luminous transmission are defined as follows:

Photopic luminous transmission Tv

$$Tv = (1/k) \int_{380}^{780} Te\ E_e\ V_e\ d_e$$

$$\text{where } k = \int_{380}^{780} Ee\ V_e\ d_e$$

T_e = spectral transmission of material at wavelength
E_e = relative spectral irradiance of source at wavelength
V_e = photopic luminous efficiency at wavelength

Scotopic luminous transmission Tv'

$$Tv' = (1/k') \int_{380}^{780} Te\ E_e\ V_e\ d_e$$

$$\text{where } k' = \int_{380}^{780} Te\ V_e\ d_e$$

V_e' = scotopic luminous efficiency at wavelength

4.5.4.3.2 Method of calculation. The photopic and scotopic luminous transmittance will be calculated for CIE Source C and the P43 phosphor. The photopic and scotopic luminous efficiency function are given in table VI. The relative spectral distribution of the CIE Source C and the P43 phosphor are given in table IV and table V respectively. The spectral transmission of the end item

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must be measured with a spectrophotometer and the integrals computer. A 2 nm increment for the integrals is recommended particularly if emission (P43 phosphor) absorption/rejection bands are narrow.

4.5.4.4 P43 transmittance. The spectral transmittance of the laser protective lens shall be measured at 10 nm intervals in the wavelength range of 380 nm to 780 nm with a spectrophotmeter. The spectral transmittance data and methods of calculation are identical to that required in 4.5.4.3 except the spectral weighing factors for unfiltered P43 emission as shown in table IV are used rather than Illuminant C. The photopic eye response weighing factors shall be used. The calculation may be done concurrently with the use of a computer or programnable calculator.

TABLE IV. Relative spectral emission of P43 Phosphor

Wavelength	P43 Phosphor	Wavelength	P43 Phosphor
380	0.083	558	0.036
382	0.125	560	0.023
384	0.139	565	0.009
386	0.109	570	0.000
388	0.051	575	0.000
390	0.021	580	0.012
395	0.000	582	0.028
400	0.000	584	0.061
405	0.000	586	0.104
410	0.008	588	0.160
412	0.031	590	0.199
414	0.068	595	0.084
416	0.118	600	0.009
418	0.143	605	0.000
420	0.102	610	0.000
425	0.011	616	0.010
430	0.007	618	0.025
435	0.039	620	0.059
440	0.084	622	0.115
445	0.017	624	0.136
450	0.000	626	0.094
455	0.009	628	0.039
460	0.021	630	0.017
465	0.000	635	0.000
470	0.006	640	0.006
475	0.021	645	0.010
480	0.012	650	0.005
485	0.034	655	0.011
490	0.160	660	0.005
495	0.173	665	0.011
500	0.036	670	0.023
505	0.010	675	0.019
510	0.000	680	0.016

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TABLE IV. Relative spectral emission of P43 Phosphor (cont'd)

Wavelength	P43 Phosphor	Wavelength	P43 Phosphor
520	0.000	682	0.021
525	0.000	685	0.014
530	0.000	690	0.000
535	0.000	700	0.000
540	0.080	710	0.000
542	0.193	720	0.000
546	0.669	730	0.000
548	0.974	740	0.000
550	0.593	750	0.000
552	0.463	760	0.000
554	0.238	770	0.000

TABLE V. Relative spectral distribution of a CIE C Source

Wavelength	CIE C Source	Wavelength	CIE C Source
380	33.00	585	95.43
385	39.92	590	93.20
390	47.40	595	91.22
395	55.17	600	89.70
400	63.30	605	88.83
405	71.81	610	88.40
410	80.60	615	88.19
415	89.63	620	88.10
420	98.10	625	88.06
425	105.8	630	88.00
430	112.4	635	87.86
435	117.75	640	87.80
440	121.50	645	87.99
445	123.45	650	88.20
450	124.00	655	88.20
455	123.60	660	87.90
460	123.10	665	87.22
465	123.30	670	86.30
470	123.80	675	85.30
475	124.00	680	84.00
480	123.90	685	82.21
485	122.92	690	88.20
490	120.70	695	78.24
495	116.90	700	76.30
500	112.10	705	74.36
505	106.98	710	72.40
510	102.30	715	70.40
515	98.91	720	68.30

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names) as applicable.

TABLE V. Relative spectral distribution of a CIE C Source (cont'd)

Wavelength	CIE C Source	Wavelength	CIE C Source
520	96.90	725	66.30
525	96.78	730	64.40
530	98.00	735	62.80
535	99.94	740	61.50
540	102.10	745	60.20
545	103.95	750	59.20
550	105.20	755	58.50
555	105.67	760	58.10
560	105.30	765	58.00
565	104.11	770	58.20
570	102.30	775	00.00
575	100.15	780	00.00
580	97.80		

TABLE VI. Photopic and scotopic luminous efficiency function

Wavelength	Photopic	Scotopic
380	.0000	.00059
385	.0001	.00111
390	.0001	.00221
395	.0002	.00453
400	.0004	.00929
405	.0006	.0185
410	.0012	.03484
415	.0022	.0604
420	.0040	.0966
425	.0073	.1436
430	.0116	.1998
435	.0168	.2625
440	.0230	.3281
445	.0298	.3931
450	.0380	.4550
455	.0480	.5129
460	.0600	.5672
465	.0739	.6205
470	.0910	.6756
475	.1126	.7337
480	.1390	.7930
485	.1693	.8509
490	.2080	.9043
495	.2586	.9491

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TABLE VI. Photopic and scotopic luminous efficiency function (cont'd)

Wavelength	Photopic	Scotopic
500	.3230	.9817
505	.4073	.9984
510	.5030	.9966
515	.6082	.9750
520	.7100	.9352
525	.7932	.8796
530	.8620	.8110
535	.9149	.7332
540	.9540	.6497
545	.9803	.5644
550	.9950	.4808
555	1.0002	.4015
560	.9950	.3288
565	.9786	.2639
570	.9520	.2076
575	.9154	.1602
580	.8700	.1212
585	.8163	.0899
590	.7570	.0655
595	.6949	.0469
600	.6310	.03325
605	.5668	.02312
610	.5030	.01593
615	.4412	.01088
620	.3810	.00737
625	.3210	.00497
630	.2650	.003335
635	.2170	.002235
640	.1750	.001497
645	.1382	.001005
650	.1070	.000677
655	.0816	.000459
660	.0610	.0003129
665	.0446	.0002146
670	.0320	.000148
675	.0232	.0001026
680	.0170	.0000716
685	.0119	.0000502
690	.0082	3.533E-05
695	.0057	2.502E-05

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TABLE VI. Photopic and scotopic luminous efficiency function (cont'd)

Wavelength	Photopic	Scotopic
700	.0041	1.78E-05
705	.0029	1.273E-05
710	.0021	9.14E-06
715	.0015	6.60E-06
720	.0010	4.78E-06
725	.0007	3.482E-06
730	.0005	2.546E-06
735	.0004	1.87E-06
740	.0003	1.379E-06
745	.0002	1.022E-06
750	.0001	7.6E-07
755	.0001	5.67E-07
760	0	4.25E-07
765	0	3.2E-07
770	0	2.41E-07
775	0	1.83E-07
780	0	1.39E-07

4.5.5 Environmental tests.

4.5.5.1 Chemical resistance. The surface shall be exposed to the specified chemicals for a 24-hour period. The chemical may be contained by sealing an O-ring to the surface using silicone grease. The O-ring shall be filled with the chemical and left for a 24-hour period. At the end of the test period, the surface shall be cleaned and inspected for visible damage and optical distortion. In addition, for those pieces which are required to provide laser protection, the luminous transmittance and the optical density shall be measured to ensure that the basic product function has not been compromised. The sample must still meet the requirements of 3.2.2.2.

4.5.5.2 Temperature. The sample shall be tested according to methods described in 4.5.4.1 and 4.5.4.3 for optical density and luminous transmittance, respectively, and shall be tested according to ANSI z80.1-1979 for optical distortion following exposure for 72 hours at 160°F and 72 hours at -60°F within ± 3°F throughout the entire period.

4.5.5.3 Solar radiation. The samples shall be tested according to the method described in MIL-STD-810. Following this exposure the lens shall be tested for compliance with the optical density and luminous transmittance requirements as specified in 3.2.2.2.

4.5.5.4 Humidity. The sample shall be subjected to a combined total of 240 hours of cycles 4 and 5 as detained in MIL-STD-810. Each cycle last 24 hours, with the extreme heat/humidity conditions as follows:

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Cycle 4: Maximum temperature 160°F, maximum humidity 80 percent at 90°F
Cycle 5: Maximum temperature 145°F, maximum humidity 70 percent at 90°F

Following this exposure, the lens shall be tested for compliance with the optical distortion, luminous transmittance, and optical density requirements as specified in 3.2.2.2.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Unit pack.

5.1.1.1 Goggles. Each pair of goggles shall be unit packed in a plastic bag conforming to A-A-50083.

5.1.1.2 Lens. Each lens shall be unit packed in a cotton flannel bag. The bag shall measure 5 by 8 ± 1/8 inches. A paper label, as described in 5.4.1.3 shall be inserted in each bag. The bag shall be closed by folding excess flannel towards the concave side of the lens.

5.1.2 Intermediate pack.

5.1.2.1 Goggles. Each pair of goggles, fitted with a class 3 lens, and one class 4 lens, unit packed as specified in 5.1.1, shall be packed in a metal edged paperboard box conforming to class 2, style D of PPP-B-665. When specified (see 6.2), other classes of lens may be used. Inside dimensions shall be approximately 8 inches in length, 4-1/4 inches in width, and 3-1/4 inches in depth. The exterior surface of the box shall be Olive Drab 7 in color.

5.1.2.2 Lens, when procured separately (see 6.2). Twenty lens of one class only, unit packed as specified in 5.1.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, variety SW, class domestic, minimum grade 125 of PPP-B-636. Inside dimensions of each box shall be approximately 7-1/4 inches in length, 4-1/2 inches in width, and 3-1/2 inches in depth. The box shall be closed in accordance with method I as specified in the appendix of PPP-B-636.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A packing. Twenty-four goggles or 600 lens of one type only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, grade V2s of PPP-B-636. Inside dimensions of the box for goggls shall be approximately 24 inches in length, 18 inches in width, and 8-1/4 inches in depth. Inside dimensions of the box for lens shall be approximately 24 inches in length, 15-1/2 inches in width, and 11-1/4 inches in depth. Each box shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Twenty-four goggles or 600 lens of one type only, preserved as specified in 5.1, shall be packed in a shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of the box for goggles shall be approximately 24 inches in length, 18 inches in width, and 8-1/4 inches in depth. Inside dimensions of the box for lens shall be approximately 23-3/4 inches in length, 15-1/2 inches in width, and 11-1/4 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. Goggles or lens preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), goggles or lens packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet patterns shall be number 90 for goggles and number 95 for lens in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, intermediate containers, shipping containers and unit loads shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

5.4.1 Special marking.

5.4.1.1 Intermediate pack for goggles (carrying container). The intermediate pack for goggles shall have printed on top of the lid with water insoluble black ink the following information:

STOCK NUMBER
ITEM DESCRIPTION
QUANTITY
LENSES: 1 each clear
1 each neutral gray
OR
1 each light
1 each dark
NAME AND ADDRESS OF MANUFACTURER

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PROPER USE OF LENSES. Use the neutral gray lens in bright light, the clear in dull light. DO NOT USE FOR SUN SCANNING. DO NOT SCRATCH THE LENSES. They are of plastic which is safer and lighter but will scratch if abused. When not in use, protect them against dirt or hard objects. Clean by washing off with water when possible or steaming with your breath and then wiping dry with soft cloth or toilet paper. Gasoline or hot water should not be used for cleaning.

COLD WEATHER PRECAUTIONS: If cold stiffens goggles, do not bend or force them. If they fog, hold away from face to let air in behind the lens until it clears.

WHEN NOT BEING WORN: Goggles should be kept in dry place away from sunlight.

WHEN WORN OVER EYEGLASSES: When the goggles are worn over eyeglasses, the temples of the eyeglasses are inserted into the large channels or grooves at each side of the goggles.

CAUTION: Avoid smearing insect repellent on lens, since it will attack the lens, thereby affecting visibility.

On the front panel of lid:

TO PROVIDE MAXIMUM VISION, WEAR GOGGLES AS LOW ON NOSE AS POSSIBLE.

5.4.1.2 Unit pack for spare class 3 or class 4 lens (paper envelope). The spare lens paper envelope shall have plainly printed on the front with water insoluble black ink the following information:

STOCK NUMBER
LENS, PLASTIC, GOGGLE. CLEAR (Class 3)
or
LENS, PLASTIC, GOGGLE. NEUTRAL GRAY (Class 4)
For Protection Against Excessive Light, Glare.

CARE OF LENS: The plastic lenses for the goggles are flexible, shatter-resistant and lighter than glass. However they are scratched much more easily than glass lenses.

When not in use, protect them against dirt or hard objects. To make them last longer, clean, dust and remove any fingerprints from them by washing with water or by breathing on the lenses, then wipe dry with a soft clean cloth or toilet paper, making absolutely certain that the cloth or paper is free of grit, sand or dirt. Gasoline should not be used for cleaning.

Replace a permanently marred lens by a new lens from your supply officer. Keep any lens, when not inserted in frame, flat in the carrying case to guard against warpage.

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CAUTION: Avoid smearing insect repellent on lens, since it will attack the lens, thereby affecting its visibility.

How to Interchange the Lenses in Your Goggle Frame

REMOVAL OF LENS

1. Unsnap the two fasteners
2. Remove the lens from the goggle

INSERTION OF REPLACEMENT LENS

3. Fasten the lens to the top snap
4. Insert the lens into the channel of the goggle frame and fasten the lens to the bottom snap.

5.4.1.3 Paper label for class 5 or class 6 lens. The class 5 or class 6 lens shall have a paper label insert (see 5.1.1.2) printed with water insoluble black in accordance with figure 2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The goggles are intended for use by combat vehicle crewman, are to be worn as a component of the face mask for flame, dust, wind and fragmentation protection, and can be worn with or without glasses.

6.2 Acquisition data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article samples are required (see 3.1, 4.3 and 6.3).
c. Type and class of goggles required (see 1.2).
d. Selection of classes of lenses when type I goggle is to be procured (see 3.3.3.1).
e. When lenses are to be procured separately (see 5.1.2 and 5.2).
f. Level of packing required (see 5.2).
g. Type and class of unit load (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.2.1 Carrying container. The carrying containers specified in 5.1.2.1 are of special design in order to serve a dual function. Initially, they provide physical protection to each pair of goggles as an intermediate package and later furnish safe storage for the goggles when not in use.

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article sample required shall be as specified in the contract and should consist of one goggle frame from each cavity along with one lens of each class. One goggle frame shall be completely assembled in accordance with 3.3.1 and 3.3.2. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Method of adhesion of fastener hook tape to goggle (type II) (see 3.2.8.1). The following method for bonding the fastener hook tape to the rubber goggle frame has been found to be successful in meeting the adhesion requirement for these parts:

a. Thoroughly abrade the areas of the goggle frame to be adhered, with 80 grit abrasive cloth.

b. Lightly abrade the back of the fastener hook tape parts, taking care not to damage the cloth material.

c. Clean the abraded areas of the goggle thoroughly with toluene and a lint-free cloth.

d. Clean the back of the fastener hook tape with a lint-free cloth dampened with toluene.

e. Brush apply one coat of adhesive to each of the surfaces to be bonded and allow to dry 15-20 minutes.

f. Brush apply a second coat of adhesive to each of the surfaces and combine while they are still wet. Apply pressure uniformly over all areas for 24 hours to achieve optimum adhesion.

Note: Cut-out sections of fastener loop tape may be used over the hook material to prevent hook damage under bonding pressure.

6.5 Foam pad source (see 3.2.6). As a matter of information only, a source for the foam pad is:

Merriweather Foam Latex Co.
11 Brown Street
Babertown, OH 44203

6.6 Solution for dipping ends. A suitable solution is available from:

Mehart Bostik Chemical Group
Middletown, MA 01949
Bostik Adhesive No. 7008

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6.7 Supersession data. Type I, classes 1 and 2 non-ballistic lenses and their requirements have been deleted. The Army is planning to furnish all (spectacle and non-spectacle wearers) of it's personnel ballistic eye protection.

6.8 Subject term (key word) listing).

Combat vehicle crewman
CVC
Facemask
Ballistic
Eye
Glasses
Shatterproof

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of

Custodians:

Army - GL

Preparing activity:

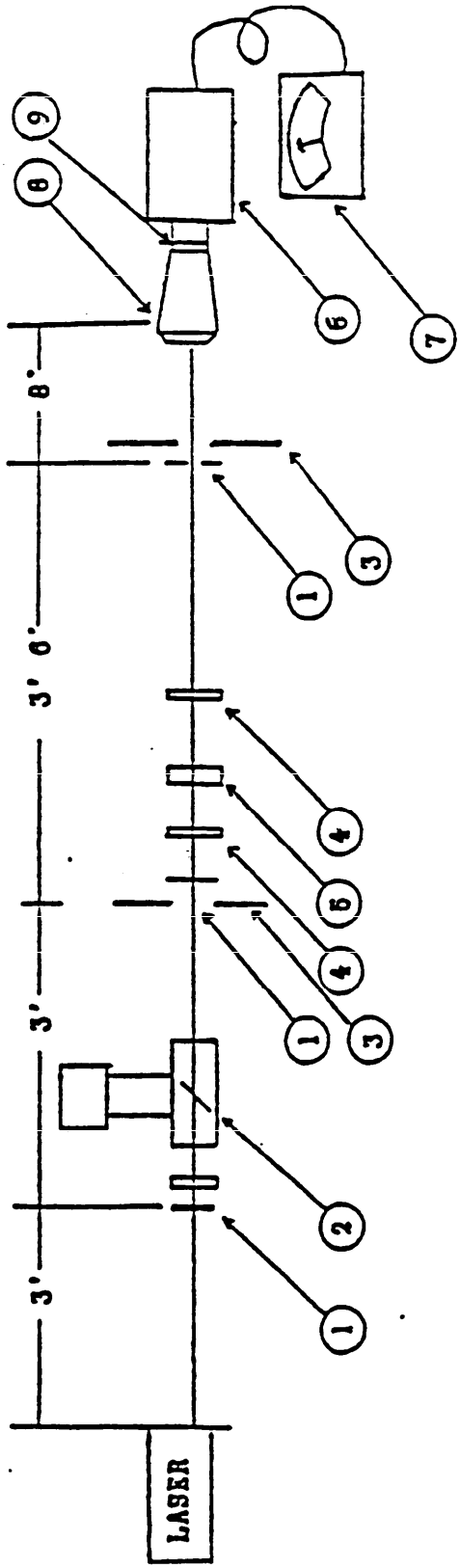
Army - GL

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FIGURE 1

RUBY LASER TEST SETUP



1. Aperatures
2. Beamsplitter and second radiometer
3. Baffles
4. Neutral density filters
5. Sample
6. Radiometer
7. Radiometer readout
8. Visible blocking filter
9. Hood and narrow band filter

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FIGURE 2 (part 1)

LASER PROTECTIVE LENSES (LPL)
FOR THE
GOGGLES, SUN, WIND AND DUST

SAFETY, CARE AND INSTRUCTION SHEET

I. GENERAL DESCRIPTION:

Laser Protective Lenses (LPL) for the Sun, Wind and Dust Goggles consist of two protective lens types:

o Green color for night use or in low light conditions. This provides protection against Ruby (694 nm) and Neodymium (1064 nm) type lasers.

o Brown color for day use. This provides protection against three types of laser; Ruby (694 nm), Neodymium (1064 nm) and Double Neodymium (532 nm). This lens may also be used as a sunglass although exposure to direct sunlight should be kept to a minimum.

Both types of lenses are polycarbonate plastic and offer Ballistic/Impact Protection. The lenses are shatter resistant however, they are not shatterproof.

These lenses are to be used in place of the standard clear lens and gray sun lens in laser threat situations. The clear and gray lenses do not provide laser protection.

Use only one lens at a time in the goggle frame. Do not attempt to nest one lens over another within the frame. This will only damage the lenses. It will not provide increased or greater protection.

II. SAFETY:

THIS ITEM MAY SAVE YOUR EYESIGHT AND YOUR LIFE!

WARNING

The Brown LPL will change the appearance of and possibly eliminate some red and/or green light sources.
The Green LPL will change the appearance of and possibly eliminate some red light sources.

CAUTION

The Green LPL will protect against Ruby and Neodymium lasers only. The Brown LPL will protect against Ruby, Neodymium, and Double Neodymium lasers only.

CAUTION

If lased, do not stare at the laser source. Some lasers have additional wavelengths that may not be filtered by these LPLs and may cause eye damage.

CAUTION

The LPLs are not intended to provide protection against bright light. Do not use to view solar eclipses, electric welding, torch welding, burning, cutting, or other potentially eye damaging light sources.

CAUTION

Laser protection levels may be reduced if used with binoculars or any magnifying sighting devices.

CAUTION

The LPLs are not to be used as a substitute for other types of laser eye protection. Protection during maintenance or servicing of specific laser systems should be as specified by the appropriate technical manual.

NOTE

The LPLs reduce ambient light levels available to the eye. With the lenses on, eyes should be allowed to adjust prior to operation during periods of dusk or dawn.

NOTE

Excessive scratching may degrade the laser protection factor of the LPLs. Clean according to instructions only. Turn in LPLs with an excessive number of scratches or any deep scratches.

NOTE

Unnecessary sunlight exposure should be avoided because excessive exposure to direct sunlight may reduce the laser protection factor of the LPLs.

III. INSTALLATION/INSPECTION/REPLACEMENT:

HOW TO INTERCHANGE THE LENSES IN YOUR GOGGLE FRAME

REMOVAL OF LENS

1. Unsnap the two fasteners located near the center of the goggles.
2. Remove the lens from the goggle.

INSERTION OF REPLACEMENT LENS

1. Fasten the lens to the top snap.
2. Insert the lens into the channel of the goggle frame and fasten the lens to the bottom snap.

o When worn over eyeglasses, temples of the eyeglasses are inserted in the side channels of the goggle frame located near the adjustment strap attachments.

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FIGURE 2 (part 2)

o Your laser protective lenses should be periodically inspected for excessive scratching, cracks, or pits.

o Replace a permanently marred or damaged lens with a new lens from your supply officer. Use care with lenses when not inserted in frame to guard against excessive warping or damage.

IV. CARE OF LENS:

o The plastic lenses for the goggles are flexible, shatter-resistant and lighter than glass. However, they are scratched much more easily than glass lenses.

o When not in use, protect them against dirt or hard objects. To make them last longer, goggles should be kept in a dry place out of direct sunlight.

o Avoid smearing insect repellent or camouflage face paint on lens, since it may attack the lens thereby affecting its visibility.

USE IT - DON'T ABUSE IT!

V. CLEANING INSTRUCTIONS

o Wash with mild soap or suitable detergent and water.

o Rinse in clean water.

o Air dry or pat dry with clean soft tissue or cloth.

o Do not use ammonia, alkaline cleaners, abrasive cleaning compounds, solvents, gasoline, bore cleaner, or hot water to clean your LPLs.

o Dust and fingerprints may be removed by breathing on lenses, then wiping dry with a soft clean cloth or toilet paper, making absolutely certain that the cloth or paper is free of grit, sand or dirt.

LASER PROTECTIVE LENS (LPL) SYSTEM

Record of Lase/Hit

Name_____

Rank_____

Unit_____

Date of Issue_____

Date of Hit_____

Duty Performed When Hit_____

Source of Laser_____

Estimated Range_____

Was Injury Sustained? YES NO

Continued to Perform Mission? YES NO

Describe Incident_____

_____

_____

_____

This record should be turned in to your Unit Commander.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-0043914E(GL)	1990 November 26

3. DOCUMENT TITLE
GOGGLES, SUN, WIND AND DUST: 1974

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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